# Segmenting Tumor Images to Study Immune Cells in Endometrial Cancer

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01

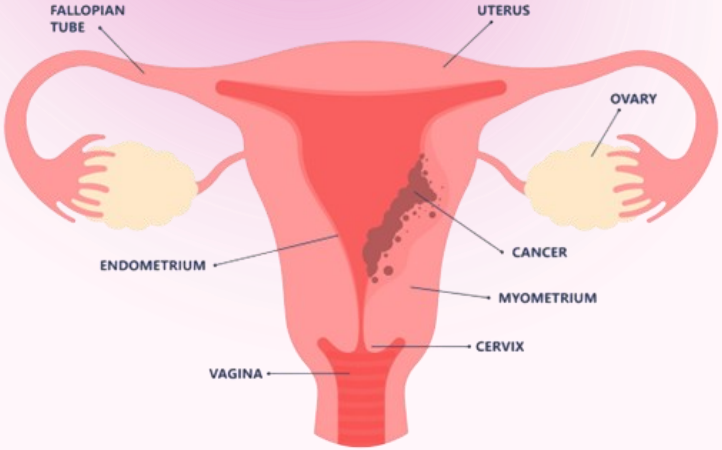
# Introduction & Background

# Clinical Context

- **Growing Health Challenge**
  **Endometrial Cancer** as the most common gynecological cancer in developed nations

- **Treatment Limitation**
  Advanced stages face poor long-term survival with chemotherapy

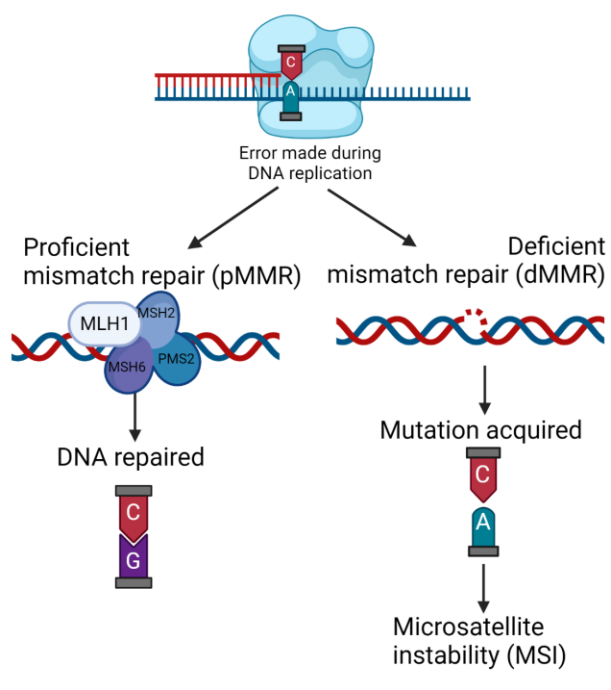
- **Immunotherapy Potential**
  High immunogenicity in specific subtypes makes them susceptible to Immune Checkpoint Inhibitors (ICIs)




https://www.vejthani.com/wp-content/uploads/2022/06/Endometrial-cancer-1.jpg

# Importance of Microsatellite Instability (MSI)

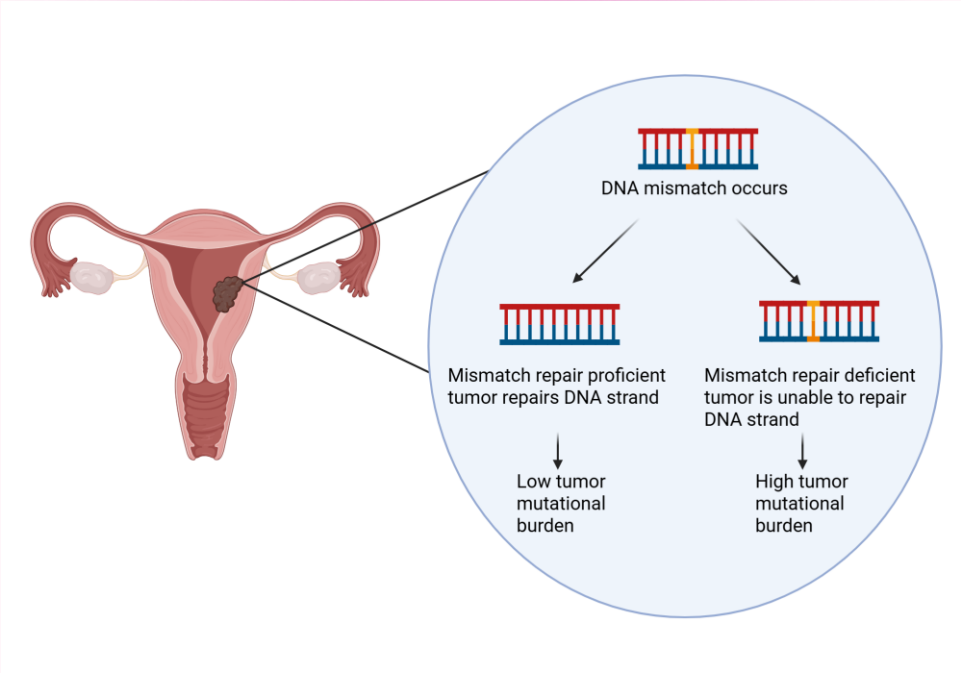
- Microsatellites are short repetitive DNA stretches
- Highly prone to replication errors
- Mismatch Repair (MMR) corrects replications errors
- Deficient Mismatch Repair (dMMR) fails to recognize these errors
- Microsatellite change length (instability)
- Mutations accumulate → MSI-H




*DNA mismatch repair deficiency*

# Importance of Tumor Mutational Burden (TMB)

- Defined as total number of somatic aquired mutations
- Microsatellite Stable (MSS) tumors typically have **Low TMB**
- **TMB-H** has many mutations, MSS/TMB-H caused by proofreading errors (DNA Polymerase Epsilon (POLE))
- High TMB generates more **neoantigens**, making the tumor more **immunogenic**




https://www.biorender.com/citation/tumor-mutational-burden-is-higher-is-mmrd-endometrial-tumors

# Molecular Subtypes

| Biomarker | Profile | Behavior | Mutation Type | Mutation Count |
|---|---|---|---|---|
| **MSI-H** | Unstable | "Hot" (Strong Immune Response) | dMMR Mismatch Repair deficiency causes instability | High<br><br>10 – 100 mut/Mb |
| **MSS/TMB-L** | Stable | "Cold" Poor Response | P53 Structural Changes rather than specific mutations | Low<br><br>< 10 mut/Mb |
| **MSS/TMB-H** | Stable | High Neoantigens | POLE Mutation Broken DNA "proofreading" leads to errors | Ultra High<br><br>> 50 mut/Mb |

# Scope of the Project

- Analyze immune cells within the MSS/TMB-H subtype
- Quantify densities and identify unique signatures
- Generate high-resolution Heatmaps of immune cell Distribution
- Develop a Machine Learning model to classify molecular subtypes

02

# Methods

# Methods

Dataset

HoVer-NeXt

Heatmap & CSV

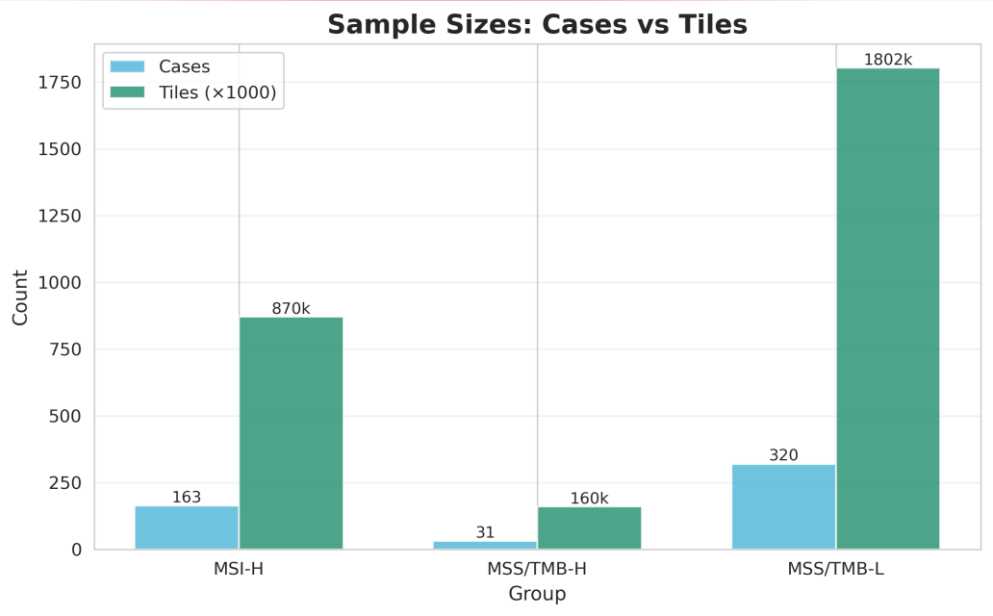
Exploratory & Statistical Analysis

Machine Learning

Unifold Manifold Approximation and Projection (UMAP)

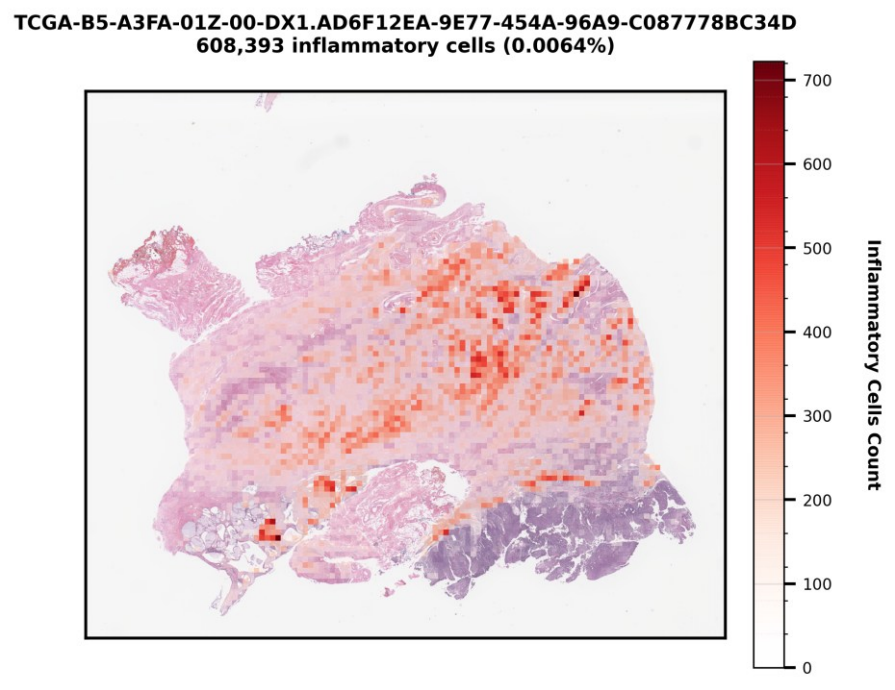
# A Highly Imbalanced Dataset

- Data from The Genome Cancer Atlas (TGCA)
- Whole Slide Images (WSIs) stained with Hematoxylin and Eosin
- 514 Cases
  **MSS/TMB-L: 320**
  **MSS/TMB-H: 31**
  **MSI-H 163**



# HoVer-NeXt

- Segmentation and Classification model based on Whole Slide Images (WSIs)
- Uses "BCB" (Boundary, Center, Background) maps
- WSIs were divided into tiles measuring 896x896 pixels
- Classification of 5 Classes: Necrosis, Inflammatory, Connective, Neoplastic, Non-Neoplastic




Spatial Heatmap of MSS/TMB-H

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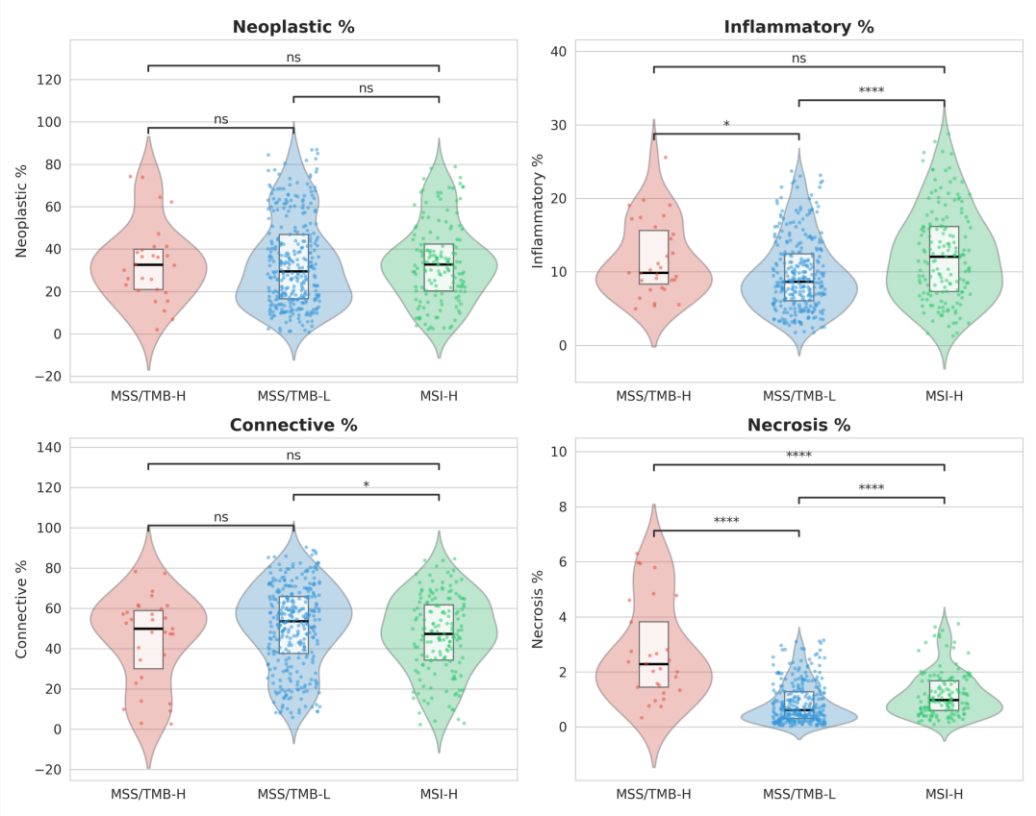
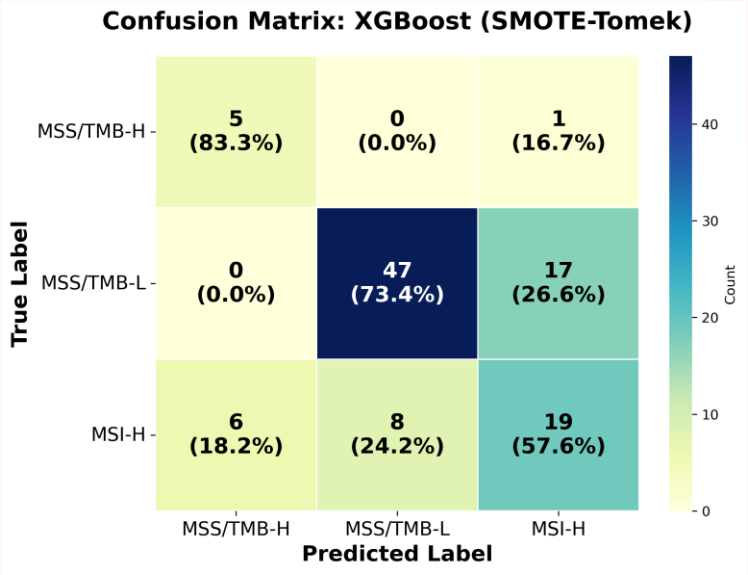
# Results

# Necrosis emerged as the Key Differentiator

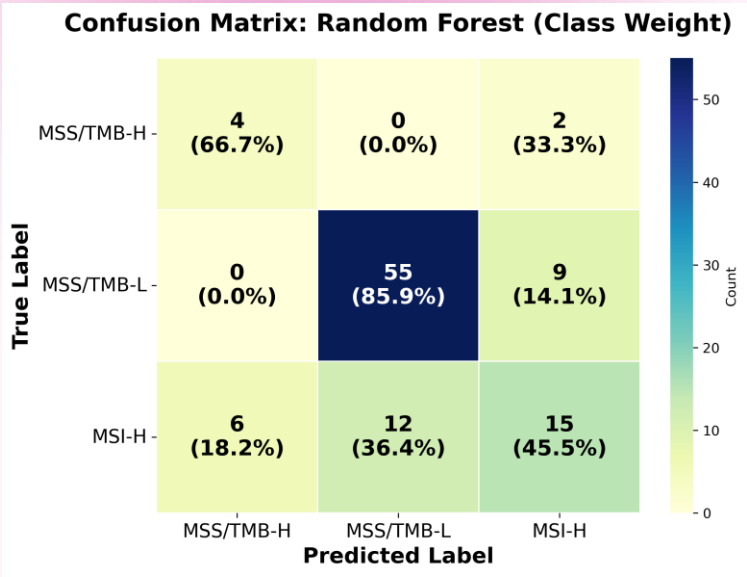
- Differences between subtypes were confirmed using Mann-Whitney U tests *($p < 0.05$)*
- Inflammatory percentages alone does not significantly differentiate MSS/TMB-H
- Necrosis emerged as the strongest differentiator with MSS/TMB-H displaying singificant elevated levels



# XGBoost Superior at Identifying Rare Subtypes



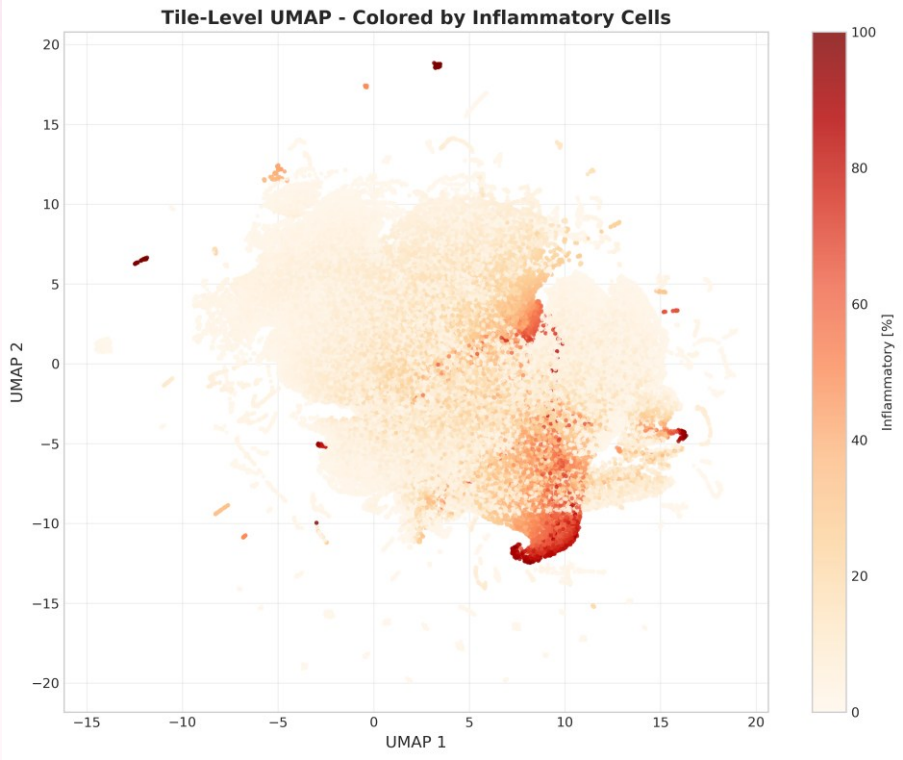

F1 = 0.64




F1 = 0.62

# Distinct Immune Niches Revealed by UMAP

- UMAP confirmed success of HoVer-NeXt
- Two prominent clusters suggests specific microenvironmental niches
- RGB composite mapping to visualize interactions



04

# Limitations & Summary

# Limitations

- Imbalanced dataset
- Only Necrotic and Inflammatory cells showed statistical significance
- Machine Learning trained on single-validation Split
- No Cell-to-Cell interactions included

# Summary

- **Necrosis** emerged as the strongest statistical differentiator, significantly elevated in MSS/TMB-H
- Machine learning models successfully predicted molecular subtypes with good accuracy
- Confusion matrices revealed shared immunogenic features between MSS/TMB-H and MSI-H
- Better prognosis for MSS/TMB-H patients

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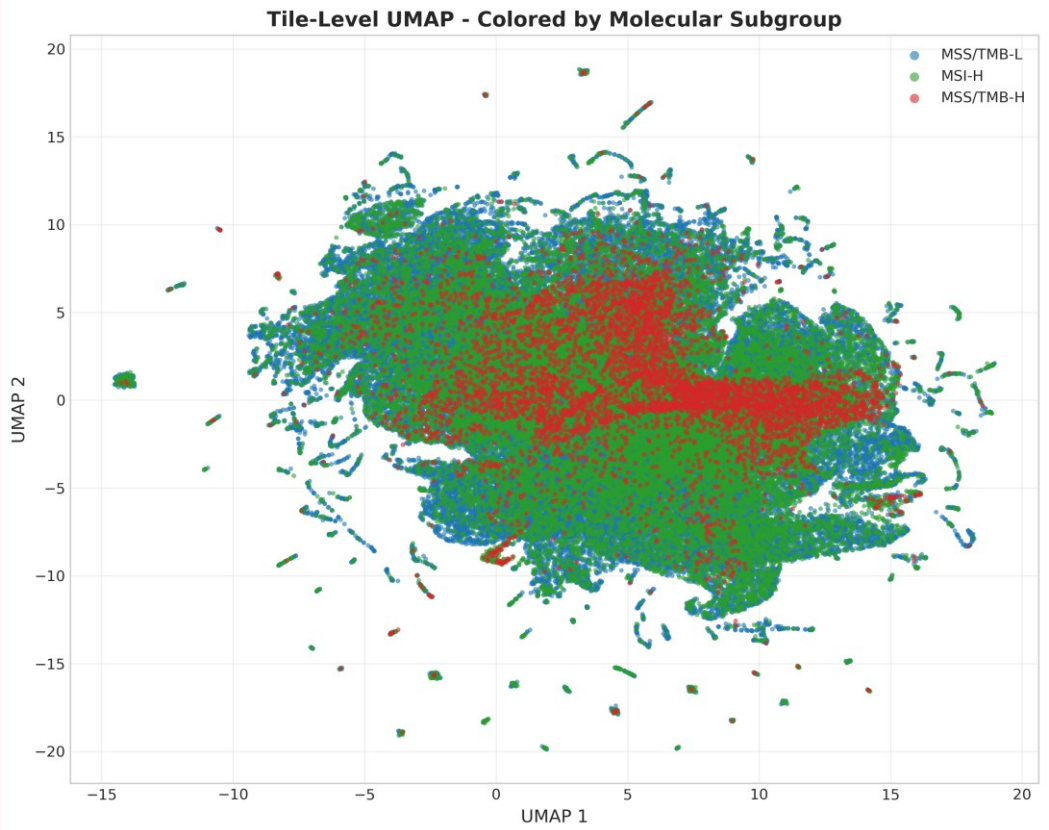
# Questions

# UMAP


Tile-Level UMAP - Colored by Molecular Subgroup
MSS/TMB-L
MSI-H
MSS/TMB-H
UMAP 2
UMAP 1
20
15
10
5
0
−5
−10
−15
−20
−15
−10
−5
0
5
10
15
20

# Sources


Albertí-Valls, M., Olave, S., Olomí, A., Macià, A., & Eritja, N. (2024). Advances in Immunotherapy for Endometrial Cancer: Insights into MMR Status and Tumor Microenvironment. *Cancers*, *16*(23), 3918. https://doi.org/10.3390/cancers16233918

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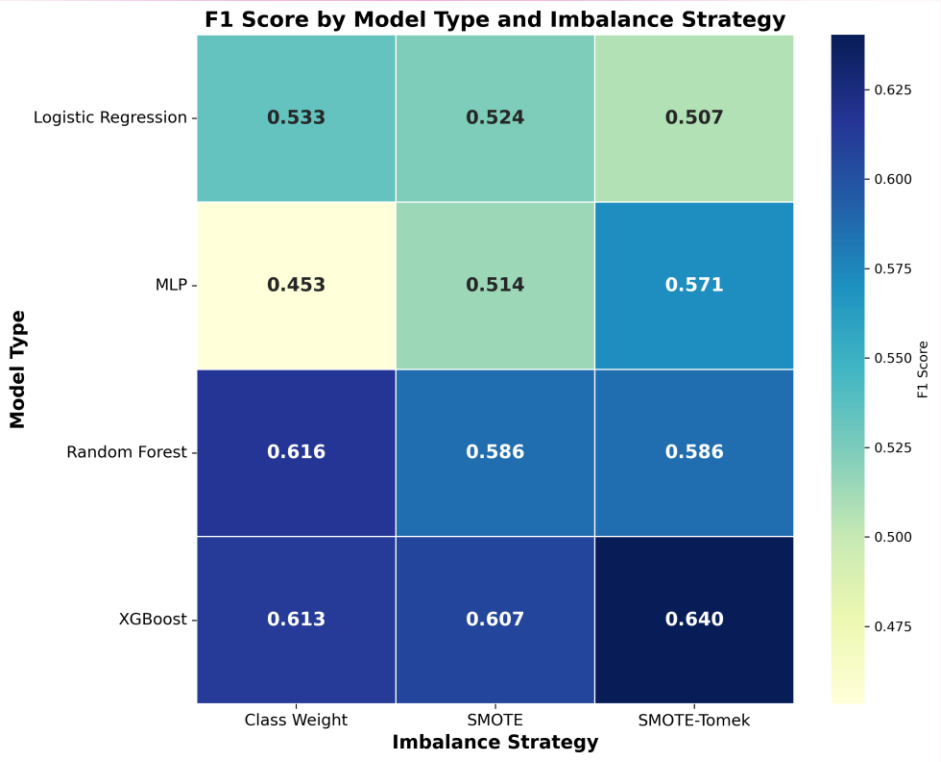
Nádorvári, M. L., Lotz, G., Kulka, J., Kiss, A., & Tímár, J. (2024). Microsatellite instability and mismatch repair protein deficiency: Equal predictive markers? *Pathology and Oncology Research*, *30*, 1611719. https://doi.org/10.3389/pore.2024.1611719

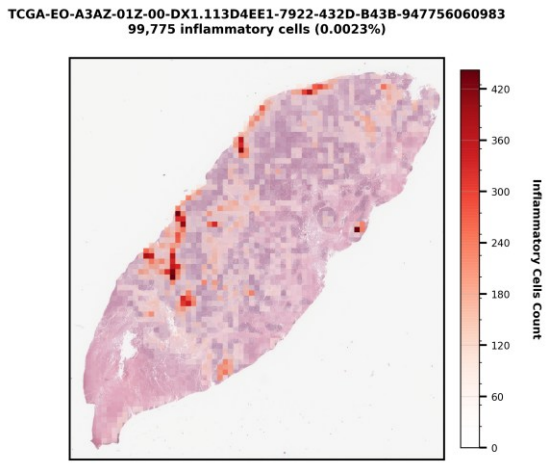
# BACKUP

# XGBoost + SMOTE-Tomek Yields the Best F1 Score

- 12 Models were Trained
- XGBoost with SMOTE-Tomek achieved the best results (F1 = 0.64)
- Random Forest with class weight was the second best (F1 = 0.62)

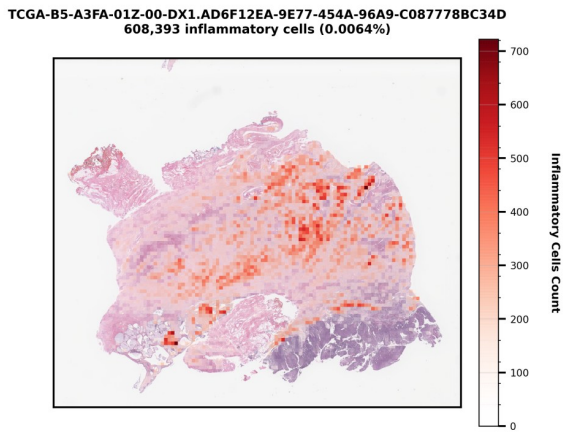


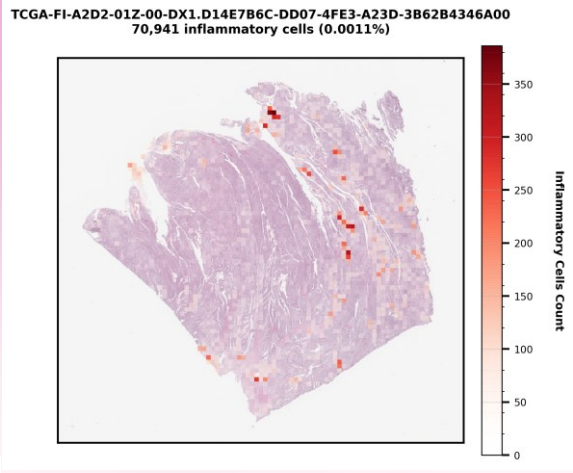
# Spatial Heatmaps




MSI-H

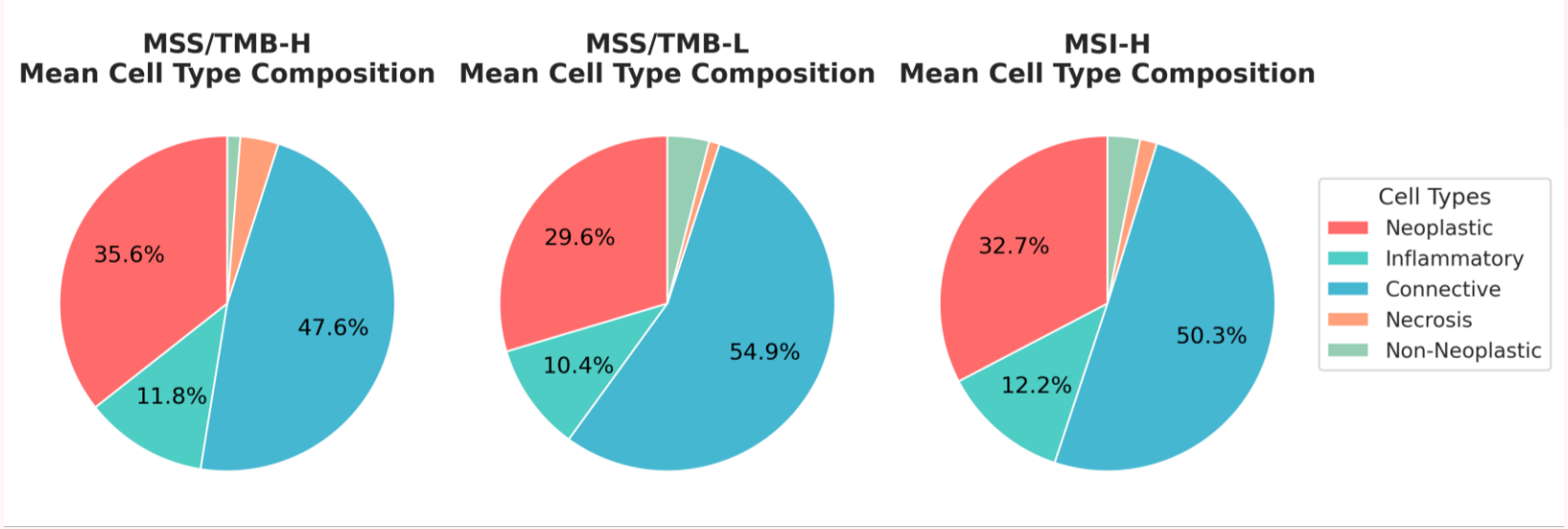
MSS/TMB-H



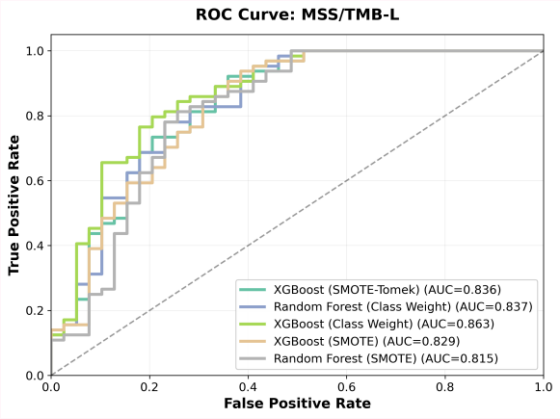



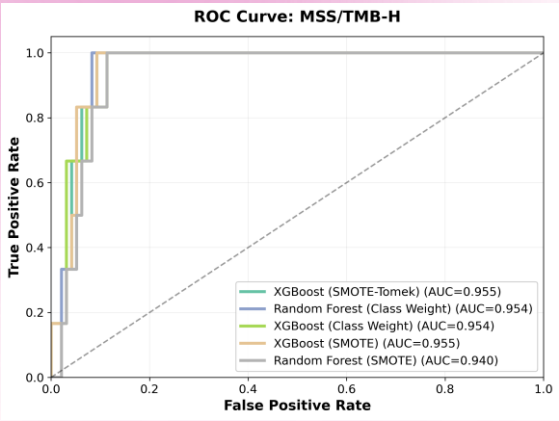

MSS/TMB-L

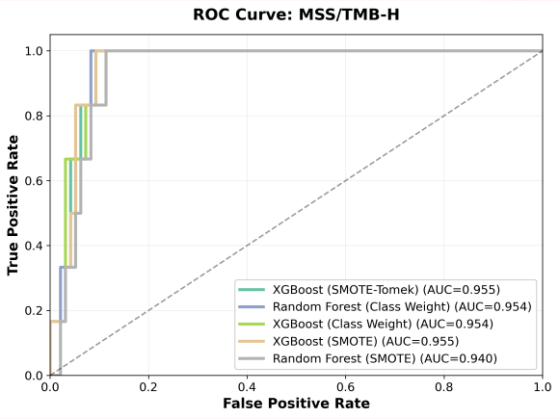
# Cellular Composition



# ROC-AUC


ROC Curve: MSS/TMB-L
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
XGBoost (SMOTE-Tomek) (AUC=0.836)
Random Forest (Class Weight) (AUC=0.837)
XGBoost (Class Weight) (AUC=0.863)
XGBoost (SMOTE) (AUC=0.829)
Random Forest (SMOTE) (AUC=0.815)


ROC Curve: MSS/TMB-H
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
XGBoost (SMOTE-Tomek) (AUC=0.955)
Random Forest (Class Weight) (AUC=0.954)
XGBoost (Class Weight) (AUC=0.954)
XGBoost (SMOTE) (AUC=0.955)
Random Forest (SMOTE) (AUC=0.940)


ROC Curve: MSS/TMB-H
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
XGBoost (SMOTE-Tomek) (AUC=0.955)
Random Forest (Class Weight) (AUC=0.954)
XGBoost (Class Weight) (AUC=0.954)
XGBoost (SMOTE) (AUC=0.955)
Random Forest (SMOTE) (AUC=0.940)